and maintain the value zero, is incompatible with quantum mechanics. *A field's value can jitter around the value zero but it can't be uniformly zero throughout a region for* more than a brief moment.[3] In technical language, physicists say that fields undergo *vacuum fluctuations.*

The random nature of vacuum field fluctuations ensures that in all but the most microscopic of regions, there are as many "up" jitters as "down" and hence they average out to zero, much as a marble surface appears perfectly smooth to the naked eye even though an electron microscope reveals that it's jagged on minuscule scales. Nevertheless, even though we can't see them directly, more than half a century ago the reality of quantum field jitters, even in empty space, was conclusively established through a simple yet profound discovery.

In 1948, the Dutch physicist Hendrik Casimir figured out how vacuum fluctuations of the electromagnetic field could be experimentally detected. Quantum theory says that the jitters of the electromagnetic field in empty space will take on a variety of shapes, as illustrated in Figure 12.1a. Casimir's breakthrough was to realize that by placing two ordinary metal plates in an otherwise empty region, as in Figure 12.1b, he could induce a subtle modification to these vacuum field jitters. Namely, the quantum equations show that in the region between the plates there will be fewer fluctuations (only those electromagnetic field fluctuations whose values vanish at the location of each plate are allowed). Casimir analyzed the implications of this reduction in field jitters and found something extraordinary. Much as a reduction in the amount of air in a region creates a pressure imbalance (for example, at high altitude you can feel the thinner air exerting less pressure on the outside of your eardrums), the reduction in quantum field jitters between the plates also yields a pressure imbalance: the quantum field jitters between the plates become a bit weaker than those outside the plates, and this imbalance *drives the plates toward each other.*